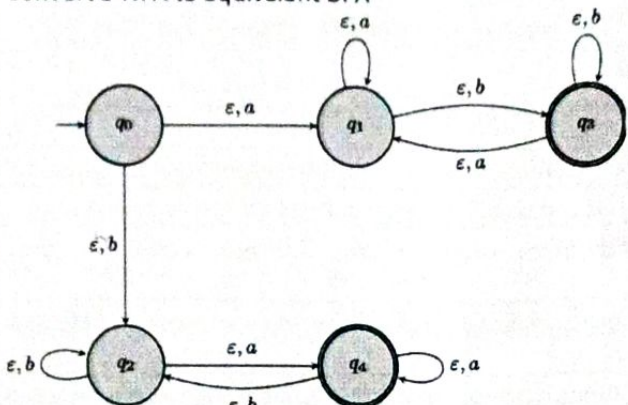
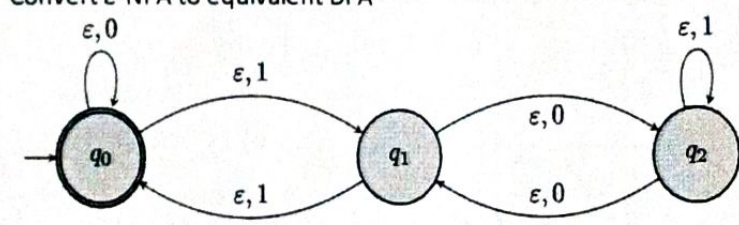
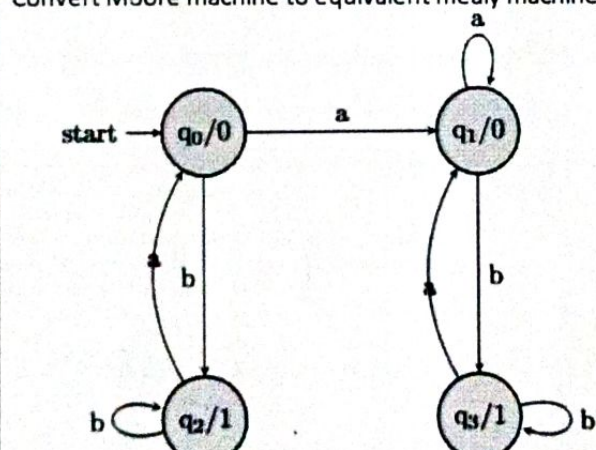


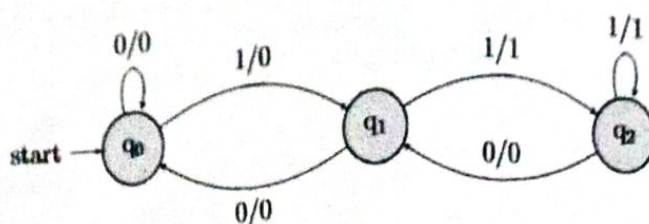
	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Information and Technology In-Semester Examination-II Academic Year: 2024-25 (Semester - I)		
Class: TY.B.Tech (IT)	Subject: Theory of Computation	Subject Code: PCC-IT307	
Date: 14/10/2025	Time: 11:00 to 11:45	Max. Marks: 20	

Note: Attempt all questions.

Draw neat diagrams wherever necessary.

Q.1	Convert ϵ-NFA to equivalent DFA 	5	CO1	BT6
Q.2	Convert ϵ-NFA to equivalent DFA 	5	CO1	BT6
Q.3	Convert Moore machine to equivalent mealy machine. 	5	CO2	BT6

Q.4	Convert mealy machine to equivalent Moore machine.	5	CO2	BT6
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== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Design deterministic and non-deterministic finite automata for regular languages.
CO 2	Construct regular expressions and context-free grammars for simple languages.
CO 3	Analyze the equivalence of different computational models such as DFA, NFA, and regular expressions.
CO 4	Apply pushdown automata and context-free grammars to parse programming language constructs.
CO 5	Explain the significance of Turing machines, decidability, and complexity in the context of computational theory.



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded
Academic Year: 2025-2026
ISE-II Examination
Class: TY (IT)

Date: 15/10/25
Time: 11:00-11:45

Subject: Introduction to Machine Learning using Python
Code: PEC-IT303

Max Marks: 20

Notes:

1. All Questions are compulsory
2. Use of a non-programmable calculator is permitted.

Question		Marks	CO	BT																								
1	A small dataset contains the following 2-dimensional points belonging to two classes: <table border="1"><thead><tr><th>Point</th><th>X</th><th>Y</th><th>Class</th></tr></thead><tbody><tr><td>P1</td><td>1</td><td>2</td><td>A</td></tr><tr><td>P2</td><td>2</td><td>3</td><td>A</td></tr><tr><td>P3</td><td>3</td><td>3</td><td>B</td></tr><tr><td>P4</td><td>6</td><td>5</td><td>B</td></tr><tr><td>P5</td><td>7</td><td>7</td><td>B</td></tr></tbody></table> A new data point $Q = (3, 4)$ is to be classified using the K-Nearest Neighbors (KNN) algorithm. a) Use Euclidean distance to calculate the distance between Q and each point in the dataset. (4 marks) b) Using $K = 3$, determine the class of Q. Show your calculations. (4 marks) c) Explain why KNN is called a lazy learning algorithm. (2 marks)	Point	X	Y	Class	P1	1	2	A	P2	2	3	A	P3	3	3	B	P4	6	5	B	P5	7	7	B	[10]	IT303.2	BT5
Point	X	Y	Class																									
P1	1	2	A																									
P2	2	3	A																									
P3	3	3	B																									
P4	6	5	B																									
P5	7	7	B																									
2	A dataset contains information about whether students pass or fail based on two features: Hours Studied and Attendance. The dataset is as follows: <table border="1"><thead><tr><th>Student</th><th>Hours Studied</th><th>Attendance</th><th>Result</th></tr></thead><tbody><tr><td>S1</td><td>2</td><td>Low</td><td>Fail</td></tr><tr><td>S2</td><td>4</td><td>High</td><td>Pass</td></tr><tr><td>S3</td><td>3</td><td>Medium</td><td>Pass</td></tr><tr><td>S4</td><td>5</td><td>High</td><td>Pass</td></tr><tr><td>S5</td><td>1</td><td>Low</td><td>Fail</td></tr></tbody></table> a) Draw a decision tree using the ID3 algorithm to classify whether a student will pass or fail. Show entropy and information gain calculations. (6 marks) b) Using your tree, predict the result for a student who studies 3 hours and has Low attendance. (2 marks) c) State one advantage and one disadvantage of using decision trees. (2 marks)	Student	Hours Studied	Attendance	Result	S1	2	Low	Fail	S2	4	High	Pass	S3	3	Medium	Pass	S4	5	High	Pass	S5	1	Low	Fail	[10]	IT303.3	BT2
Student	Hours Studied	Attendance	Result																									
S1	2	Low	Fail																									
S2	4	High	Pass																									
S3	3	Medium	Pass																									
S4	5	High	Pass																									
S5	1	Low	Fail																									

Course Outcomes (CO)

- IT303.1 To understand Probability and Statistics
IT303.2 To know what kind of data
IT303.3 To learn the fundamental topics related to Machine learning
IT303.4 Implementing advanced Machine Learning algorithms on the data
IT303.5 Understanding Advanced Systems

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create



SGGS Institute of Engineering and Technology, Vishnupuri, Nanded

Department of Information Technology

MDM in Data Science

ISE II Examination

Academic Year: 2025-26 (Semester-I)

Class: TY B.Tech.	Course: Introduction to Data Science and Visualization	Course Code: MDM-IS-IT301
Date: 16-10-2025	Time: 11:00 to 11:45	Max. Marks: 20

Instructions: 1. Make suitable assumptions wherever necessary. 2. All Questions are compulsory.

Q.N.	Questions	Marks	CO X	BT X
Q.1	Carefully review the following Python instructions and give answers based on how they would have resulted on a python interpreter. Provide valid justification for your answers, without it no marks will be awarded. For each of the following questions, assume: import numpy as np a. <pre>>>> arr = np.array([[[1, 2], [4, 5]], [[7, 8], [10, 11]]]) >>> print(arr[1, 1, 0])</pre> b. <pre>>>> arr = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9]) >>> newarr = arr.reshape(3, 3, 1) >>> print(newarr)</pre> c. <pre>>>> arr = np.array([[1, 2, 3, 4, 5], [6, 7, 8, 9, 10]]) >>> print(arr[0:2, 2])</pre> d. <pre>>>> arr1 = np.array([[1, 2], [3, 4]]) >>> arr2 = np.array([[5, 6], [7, 8]]) >>> arr = np.concatenate((arr1, arr2), axis=1) >>> print(arr)</pre> e. <pre>>>> print(np.array([1,2,3,4], ndim=5))</pre>	[10]	3	3
Q.2	a. Provide a tabular comparison between <i>Python list</i> and <i>NumPy ndarray</i> , highlighting at least six key differences. b. Given the <i>NumPy array</i> 'a', what will be the output of <code>a[::-1, :-1]</code> ? Explain your answer with a suitable example.	[3] [2]	1 2	2 4
Q.3	Develop a <i>Python</i> program utilizing <i>Pandas</i> to record marks for an arbitrary number of students in three subjects: <i>Physics, Chemistry, and Mathematics</i> . Ensure that all marks are validated as non-negative integers within the range of 0 to 100. The program must dynamically store student data, calculate each student's total and average marks, identify all top-performing students, and produce a summary of class averages for each subject. Write a python program with robust input validation and proper comments explaining what each line does.	[5]	2	6

"Python was created to be fun to use, and it has stayed that way throughout its evolution." - Guido Van Rossum
Creator of Python Programming language. **All The BEST!**

Course Outcomes (CO)

- CO1. Gain foundational knowledge in Data Science principles, Python components, and its applications across different sectors.
- CO2. Develop skills in the Data Analytics Process, including Exploratory Data Analysis techniques and drawing conclusions from data
- CO3. Acquire proficiency in feature generation and selection methods for extracting meaningful insights from data
- CO4. Master the principles of Data Visualization and create effective visualizations using various techniques.
- CO5. Utilize advanced visualization tools like Tableau and Power BI to present complex datasets creatively and effectively.

Bloom's Taxonomy (BT)

BT1- Remember, BT2- Understand, BT3- Apply, BT4- Analyze, BT5- Evaluate, BT6- Create



Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded
Department of Information Technology
In-Semester Examination II
Academic Year: 2025-26 (Semester – I)

Class: TY B.Tech	Subject: Database Management System	Subject Code: MDM-IT301
Date: 16.10.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note:

- i. Assume suitable data whenever necessary and state it clearly.
- ii. Mobile phones and programmable calculators are strictly prohibited.
- iii. Exchange/sharing of stationery, calculator, etc. is not allowed.
- iv. Figures to the right indicate full marks.
- v. The symbols carry the usual meaning.

Q.N.		Question	Marks	CO	BT
Q.1	a	Explain Super key, Candidate key, Primary Key and Foreign Key	5	CO1	BT2
	b	Consider a relation R (A , B , C , D , E , F , G) with the functional dependencies- A → BC BC → DE D → F CF → G Evaluate attribute closure of A i.e. A ⁺	5	CO3	BT5
Q.2		Compare Inner join, Natural join, Left Outer join, Right Outer join, Full Outer join and describe with example.	10	CO2	BT4

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyse, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	On successful completion of this course, students should be able to:
CO 1	Apply knowledge of the relational data model to design and represent real-world entities and relationships.
CO 2	Write, execute, and analyse various SQL to manage data in relational databases.
CO 3	Implement normalization techniques to improve database design, reduce redundancy, and enhance data integrity.
CO 4	Explain the importance of transactions and ACID properties for ensuring data consistency and reliability.
CO5	To understand organization of data on secondary storage

	Shri Guru Gobind Singhji Institute of Engineering and Technology, Nanded Department of Information Technology In-Semester Examination-II Academic Year: 2024-25 (Semester – I)	
Class: T.Y	Subject: E-Commerce	Subject Code: OE-IT301
Date: 17.10.2025	Time: 11:00 to 11:45	Max. Marks: 20

Note: Attempt any Four questions.

Draw neat diagrams wherever necessary.

Q.1	Explain B2C business model in detail.	5	CO2	BT2
Q.2	Summarize SWOT analysis in E-commerce.	5	CO4	BT2
Q.3	Enlist and explain different software tool for interactivity in E-commerce.	5	CO4	BT1
Q.4	Explain TCP/IP communication protocol in detail.	5	CO1	BT2
Q.5	Draw and explain Web Site Systems Development Life Cycle.	5	CO5	BT4

== ALL THE BEST ==

Bloom's Taxonomy (BT)

BT1 - Remember, BT2 - Understand, BT3 - Apply, BT4 - Analyze, BT5 - Evaluate, BT6 - Create

Course Outcomes:

	Course Outcomes
CO 1	Demonstrate knowledge of e-commerce concepts and technologies.
CO 2	Evaluate various e-commerce business strategies and models.
CO 3	Implement secure online transactions and payment gateways.
CO 4	Apply digital marketing techniques to enhance online business presence.
CO 5	Design and optimize user-friendly e-commerce websites.